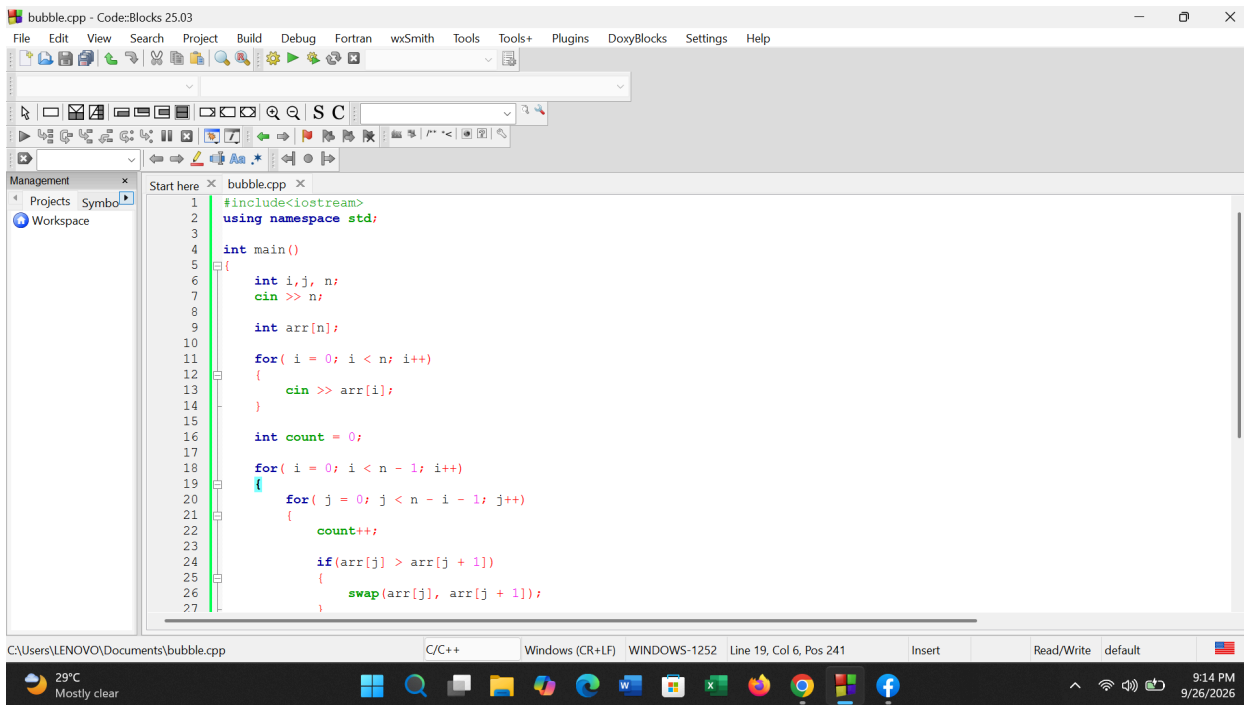
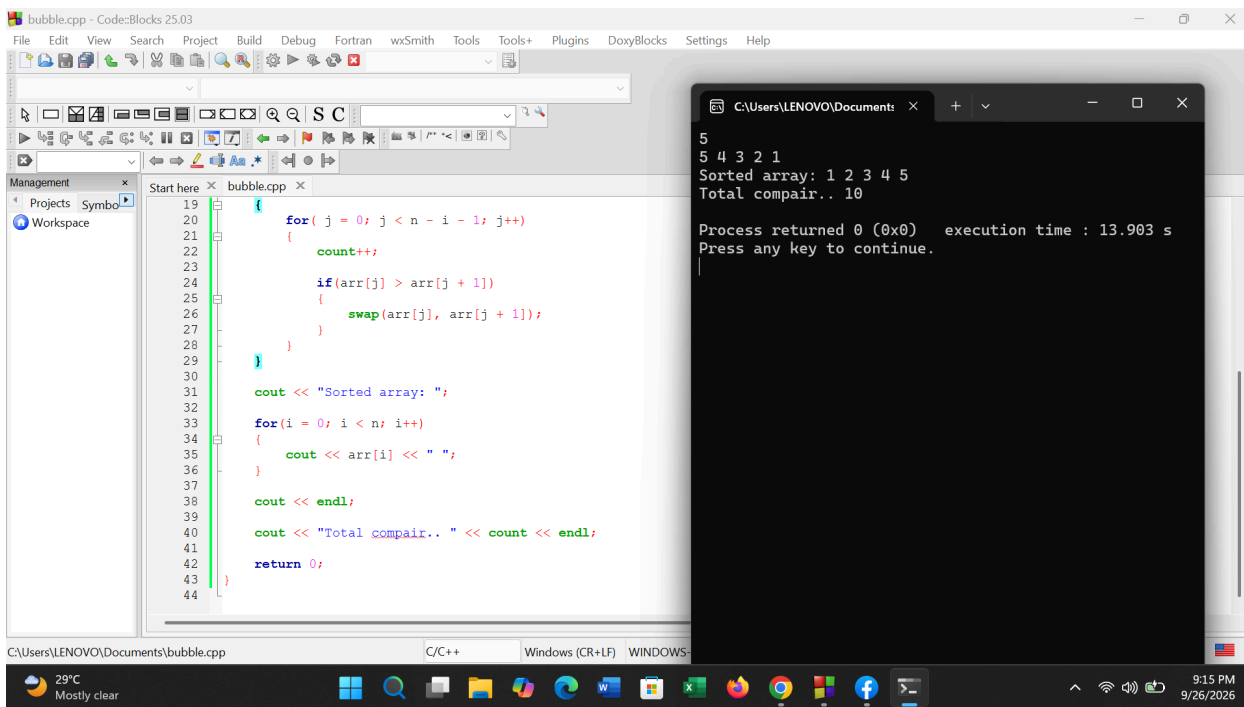


1. Bubble Sort<119>



The screenshot shows the Code::Blocks IDE with the file `bubble.cpp` open. The code defines a `main` function that takes an integer `n` and an array `arr` of size `n`. It implements the first pass of bubble sort, where it iterates through the array from index 0 to `n-1`, comparing adjacent elements and swapping them if they are in the wrong order. The `count` variable is used to track the number of comparisons.

```
1 #include<iostream>
2 using namespace std;
3
4 int main()
5 {
6     int i,j, n;
7     cin >> n;
8
9     int arr[n];
10
11     for( i = 0; i < n; i++)
12     {
13         cin >> arr[i];
14     }
15
16     int count = 0;
17
18     for( i = 0; i < n - 1; i++)
19     {
20         for( j = 0; j < n - i - 1; j++)
21         {
22             count++;
23
24             if(arr[j] > arr[j + 1])
25             {
26                 swap(arr[j], arr[j + 1]);
27             }
28         }
29     }
```



The screenshot shows the Code::Blocks IDE with the file `bubble.cpp` open. The code continues from the previous screenshot, adding the output of the sorted array and the total number of comparisons. The program's output is displayed in a terminal window, showing the input array `5 4 3 2 1`, the sorted array `1 2 3 4 5`, and the total number of comparisons `10`. The program returns `0` and the execution time is `13.903 s`.

```
19 {
20     for( j = 0; j < n - i - 1; j++)
21     {
22         count++;
23
24         if(arr[j] > arr[j + 1])
25         {
26             swap(arr[j], arr[j + 1]);
27         }
28     }
29
30     cout << "Sorted array: ";
31
32     for(i = 0; i < n; i++)
33     {
34         cout << arr[i] << " ";
35     }
36
37     cout << endl;
38
39     cout << "Total compar.. " << count << endl;
40
41     return 0;
42 }
43
44
```

Output:

```
5
5 4 3 2 1
Sorted array: 1 2 3 4 5
Total compar.. 10

Process returned 0 (0x0)   execution time : 13.903 s
Press any key to continue.
```

2 Selection Sort <119>

The screenshot shows the Code::Blocks IDE with the file 'selection.cpp' open. The code defines a main function that takes an integer 'n' and an array 'arr' of size 'n'. It initializes 'count' to 0 and 'minIndex' to 0. A loop for 'i' from 0 to n-1 is shown, with an inner loop for 'j' from i+1 to n-1. The inner loop compares 'arr[j]' with 'arr[minIndex]' and increments 'count' if 'arr[j]' is smaller. The code is at line 118, column 8.

```
1 #include <iostream>
2 using namespace std;
3
4 int main()
5 {
6     int i, j, n, count = 0, minIndex, temp;
7
8     cin >> n;
9
10    int arr[n];
11    for( i = 0; i < n; i++)
12    {
13        cin >> arr[i];
14    }
15
16    for( i = 0; i < n - 1; i++)
17    {
18        minIndex = i;
19        for( j = i + 1; j < n; j++)
20        {
21            count++;
22            if(arr[j] < arr[minIndex])
23            {
24                minIndex = j;
25            }
26        }
27    }
28    temp = arr[i];
29    arr[i] = arr[minIndex];
30    arr[minIndex] = temp;
31 }
```

The screenshot shows the Code::Blocks IDE with the file 'selection.cpp' open. The code continues from the previous screenshot, showing the inner loop for 'j' from i+1 to n-1. It compares 'arr[j]' with 'arr[minIndex]' and increments 'count' if 'arr[j]' is smaller. The code is at line 118, column 8. A terminal window is open, showing the output of the program.

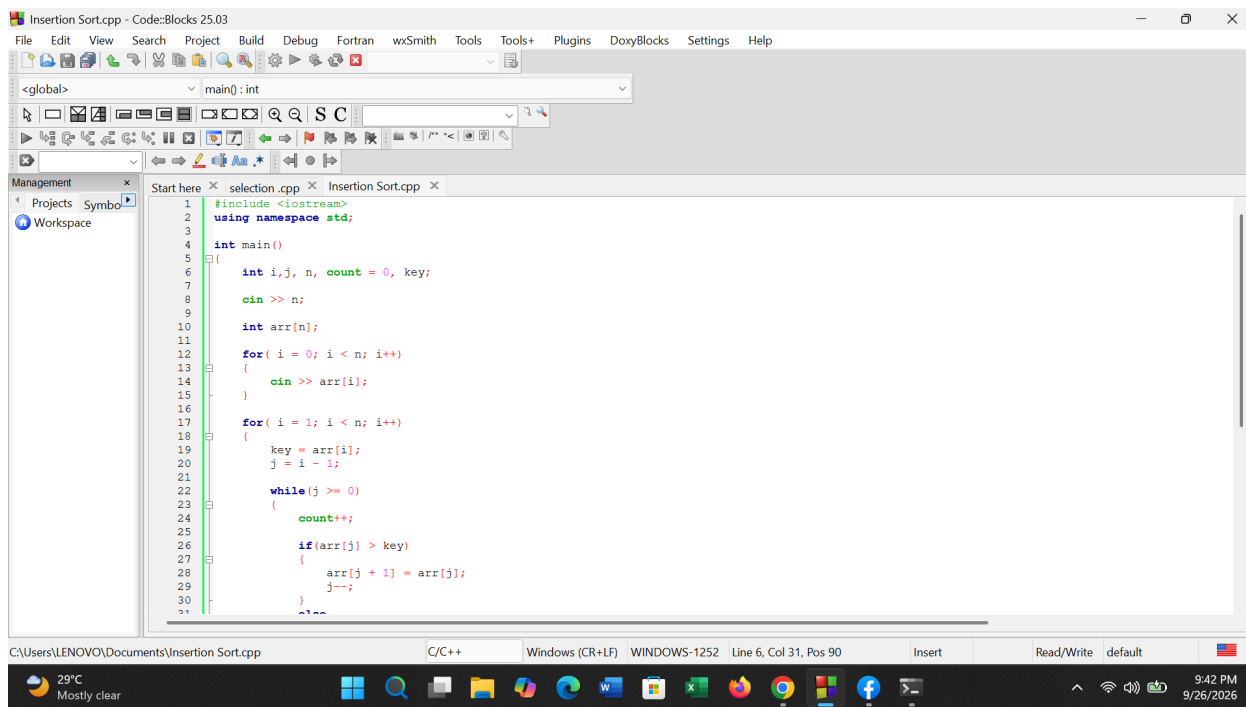
```
21     for( j = i + 1; j < n; j++)
22     {
23         count++;
24         if(arr[j] < arr[minIndex])
25         {
26             minIndex = j;
27         }
28     }
29
30     temp = arr[i];
31     arr[i] = arr[minIndex];
32     arr[minIndex] = temp;
33 }
34
35 cout << "after ... ";
36
37 for(int i = 0; i < n; i++)
38 {
39     cout << arr[i] << " ";
40 }
41
42 cout << endl;
43
44 cout << "Total compar..: " << count << endl;
45
46 return 0;
47
48
49
50 }
```

Terminal Output:

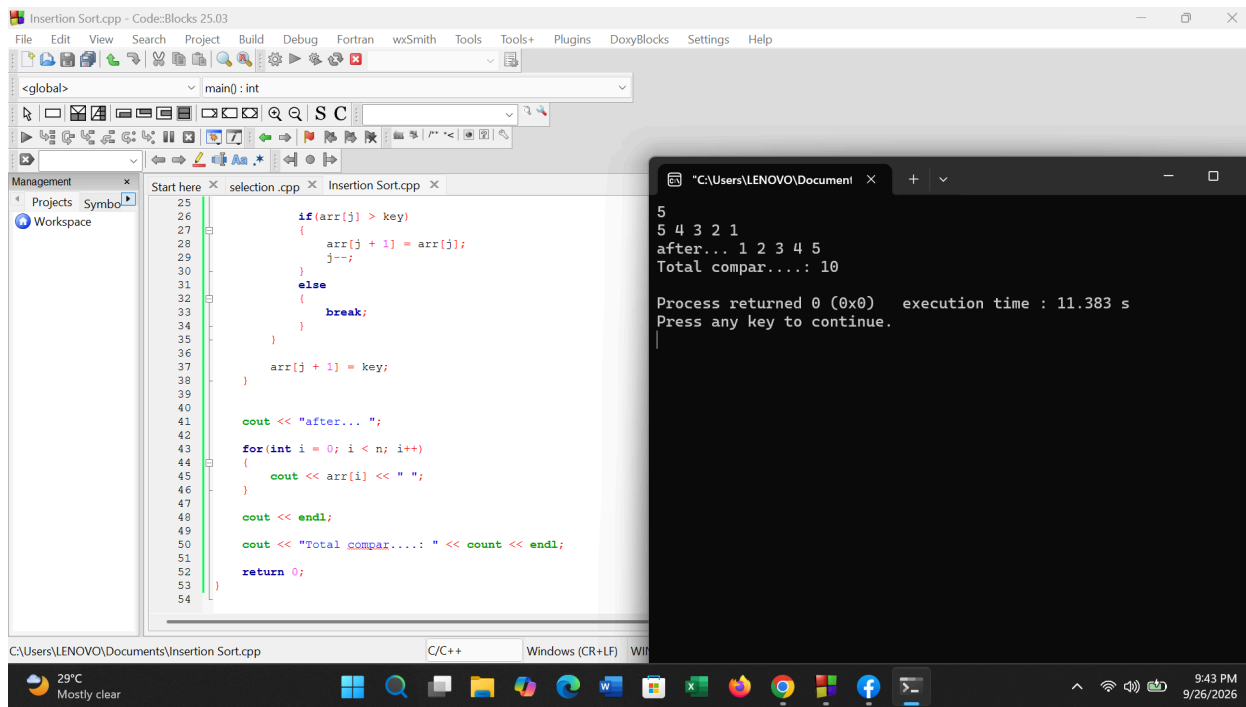
```
5
5 4 3 2 1
after ... 1 2 3 4 5
Total compar..: 10

Process returned 0 (0x0)   execution time : 6.841 s
Press any key to continue.
```

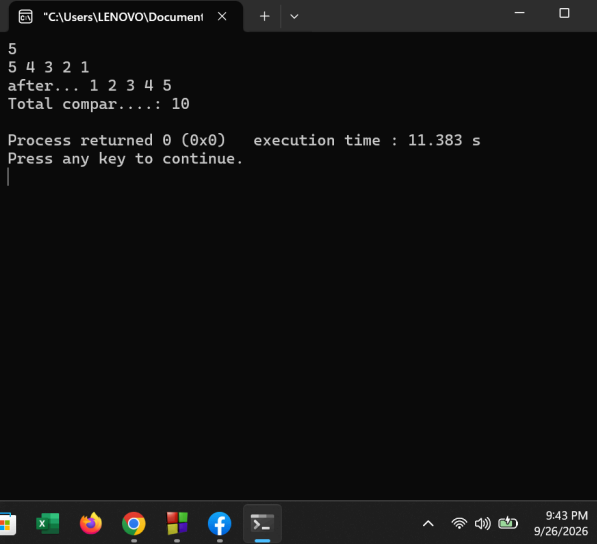
3 Insertion Sort <119>



```
1 #include <iostream>
2 using namespace std;
3
4 int main()
5 {
6     int i, j, n, count = 0, key;
7
8     cin >> n;
9
10    int arr[n];
11
12    for (i = 0; i < n; i++)
13    {
14        cin >> arr[i];
15    }
16
17    for (i = 1; i < n; i++)
18    {
19        key = arr[i];
20        j = i - 1;
21
22        while (j >= 0)
23        {
24            count++;
25
26            if (arr[j] > key)
27            {
28                arr[j + 1] = arr[j];
29                j--;
30            }
31            else
32            {
33                break;
34            }
35
36            arr[j + 1] = key;
37        }
38    }
39
40    cout << "after... ";
41
42    for (int i = 0; i < n; i++)
43    {
44        cout << arr[i] << " ";
45    }
46
47    cout << endl;
48
49    cout << "Total compar....: " << count << endl;
50
51    return 0;
52 }
```



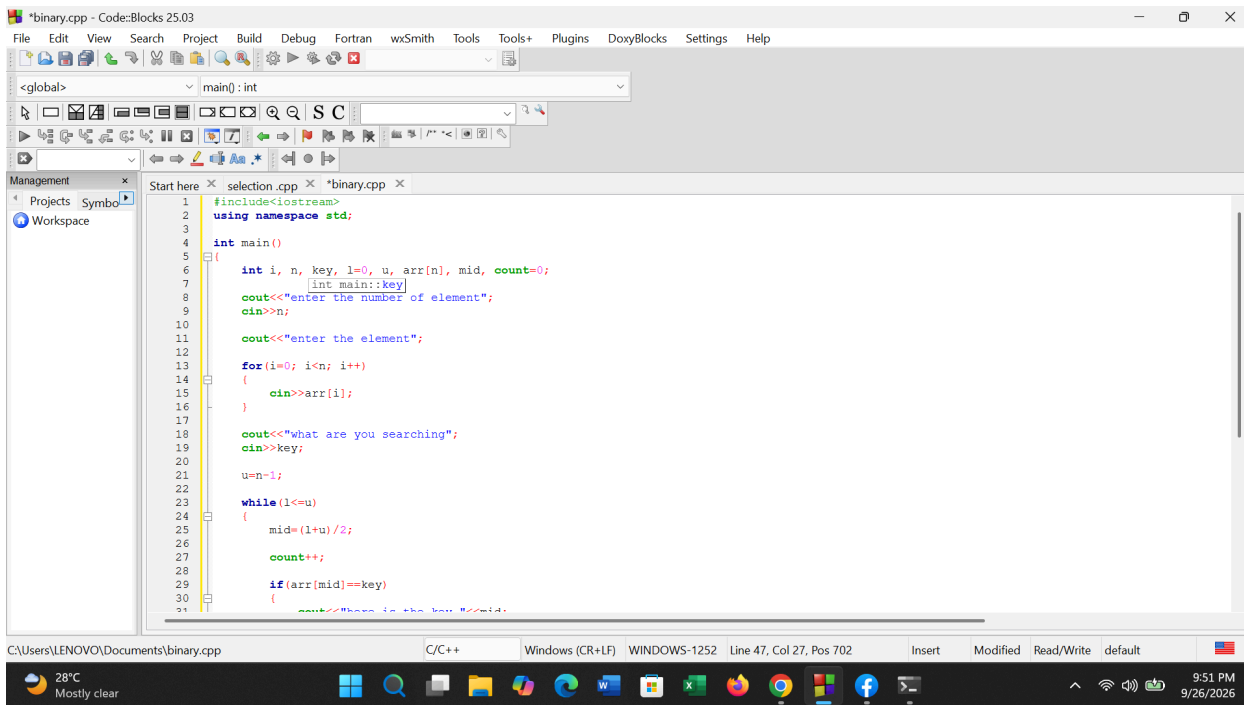
```
25
26     if (arr[j] > key)
27     {
28         arr[j + 1] = arr[j];
29         j--;
30     }
31     else
32     {
33         break;
34     }
35
36     arr[j + 1] = key;
37 }
38
39
40
41 cout << "after... ";
42
43 for (int i = 0; i < n; i++)
44 {
45     cout << arr[i] << " ";
46 }
47
48 cout << endl;
49
50 cout << "Total compar....: " << count << endl;
51
52 return 0;
53 }
```



```
5
5 4 3 2 1
after... 1 2 3 4 5
Total compar....: 10

Process returned 0 (0x0)   execution time : 11.383 s
Press any key to continue.
```

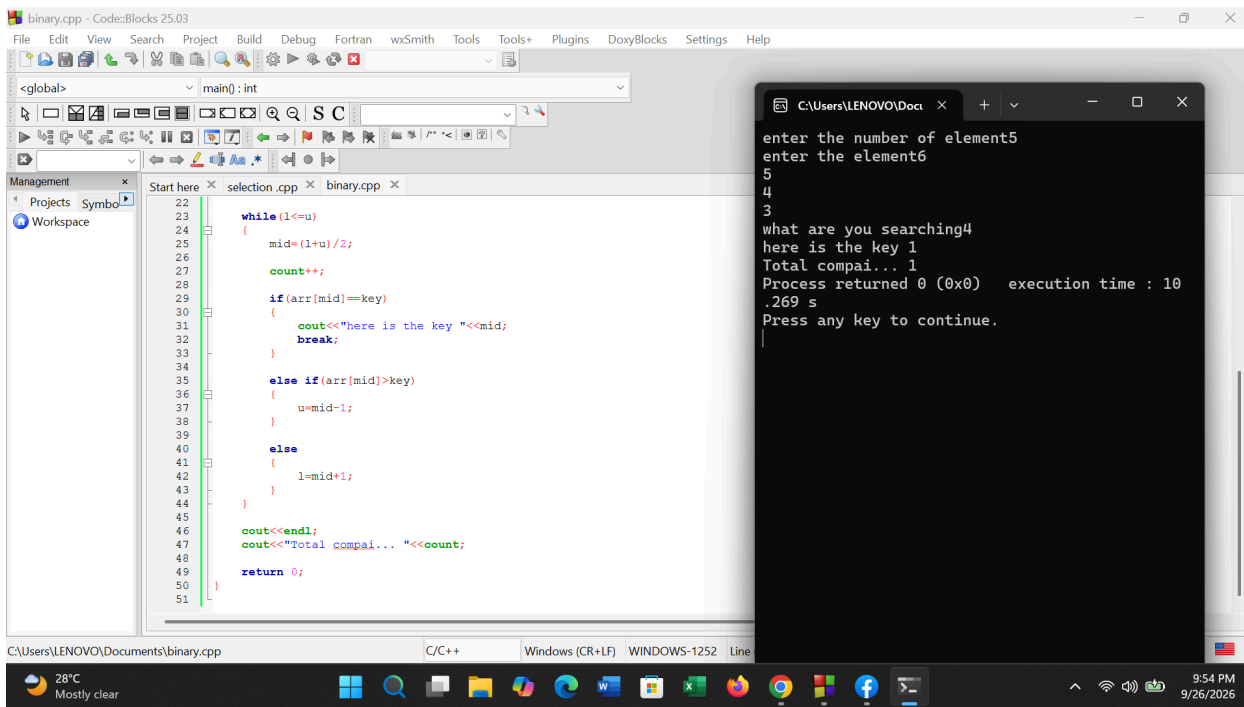
4 Binary Search <119>



The screenshot shows the Code::Blocks IDE with a C++ project named 'binary.cpp'. The code is in the 'main()' function and implements a basic binary search. It prompts the user to enter the number of elements and the elements themselves. It then prompts for a search key and enters a while loop to perform the search. The code is as follows:

```
1 #include<iostream>
2 using namespace std;
3
4 int main()
5 {
6     int i, n, key, l=0, u, arr[n], mid, count=0;
7     [int main::key]
8     cout<<"enter the number of element";
9     cin>>n;
10
11     cout<<"enter the element";
12
13     for(i=0; i<n; i++)
14     {
15         cin>>arr[i];
16     }
17
18     cout<<"what are you searching";
19     cin>>key;
20
21     u=n-1;
22
23     while (l<=u)
24     {
25         mid=(l+u)/2;
26
27         count++;
28
29         if(arr[mid]==key)
30         {
31             cout<<"here is the key "<<mid;
```

The status bar at the bottom indicates the file path is 'C:\Users\LENOVO\Documents\binary.cpp', the language is 'C/C++', and the window title is 'Windows (CR+LF) WINDOWS-1252 Line 47, Col 27, Pos 702'. The system tray shows a temperature of 28°C and the date/time as 9:51 PM on 9/26/2026.



This screenshot shows the same Code::Blocks IDE with the completed binary search code. The code includes a break statement when the key is found and calculates the total comparisons. The code is as follows:

```
22
23
24     while (l<=u)
25     {
26         mid=(l+u)/2;
27
28         count++;
29
30         if(arr[mid]==key)
31         {
32             cout<<"here is the key "<<mid;
33             break;
34         }
35
36         else if(arr[mid]>key)
37         {
38             u=mid-1;
39         }
40
41         else
42         {
43             l=mid+1;
44         }
45
46     }
47
48     cout<<"endl;
49     cout<<"Total compai... "<<count;
50
51     return 0;
```

Overlaid on the IDE is a terminal window showing the program's execution. The user input and program output are as follows:

```
enter the number of element5
enter the element6
5
4
3
what are you searching4
here is the key 1
Total compai... 1
Process returned 0 (0x0)   execution time : 10
.269 s
Press any key to continue.
```

The status bar at the bottom shows the file path 'C:\Users\LENOVO\Documents\binary.cpp', language 'C/C++', and window title 'Windows (CR+LF) WINDOWS-1252 Line'. The system tray shows a temperature of 28°C and the date/time as 9:54 PM on 9/26/2026.